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Greater Noida, Uttar Pradesh · B.Tech (CSE) — 7th Semester

Student: **Yash Mahato** ·

Enrollment: **CS-2341851**

Organisation: **Bokaro Steel Plant (SAIL)**

Project: **Python Programming and Data Analysis Using Training Datasets**

Agenda

01 — About the Organisation

Overview of Bokaro Steel Plant (SAIL) and industrial context.

02 — Internship Overview

Duration, role, training structure and focus areas.

03 — Objectives & Scope

Goals set for the training period and expected deliverables.

04 — Tools & Technologies

Python stack, libraries, environments and version control.

05 — Methodology & Workflow

Stepwise approach from data ingestion to reporting.

06 — Outcomes & Learnings

Results, technical skills gained and professional growth.

SAIL INDUSTRY OVERVIEW

About Bokaro Steel Plant (SAIL)

Bokaro Steel Plant is a major integrated steel complex of the Steel Authority of India Limited (SAIL). Established with Soviet collaboration, it includes coke ovens, blast furnaces, steel melting shops and rolling mills. The plant combines heavy industrial processes with supporting IT systems for production, inventory and quality control.

The plant regularly hosts engineering trainees, offering exposure to both shop-floor operations and digital data systems — an ideal environment for applied data analysis.

Internship Overview

Placement & Timeline

4 weeks: 07 Jul – 02 Aug 2025. Hosted at Bokaro Steel Plant (SAIL), Jharkhand.

Student Profile & Focus

B.Tech (CSE), 7th Semester. Primary focus: Python programming and analysis of trained datasets to support operational insights.

Training Structure

Combination of instructor-led sessions, hands-on labs, dataset projects and mentorship from plant engineers and IT staff.

Project Objectives

Practical Python Skills

Apply Python 3 to real industrial datasets — scripting, automation and reproducible analysis.

Data Preparation

Collect, clean and pre-process raw and trained datasets; handle nulls, duplicates and outliers.

Model Application

Use trained ML models via Scikit-learn to run predictions and validate outcomes against domain expectations.

Analysis & Reporting

Perform EDA, generate visualisations and prepare concise reports for engineering stakeholders.

Professional Development

Improve documentation, teamwork and technical communication within an industrial setting.

Tools & Technologies

Python 3

Primary language for scripting, data manipulation and model orchestration.

Pandas & NumPy

Data wrangling, transformation and numerical operations on large tabular datasets.

Matplotlib & Seaborn

Visualisation libraries for exploratory plots, trend charts and publication-ready figures.

Scikit-learn

Apply trained models, evaluate performance metrics and run cross-validation experiments.

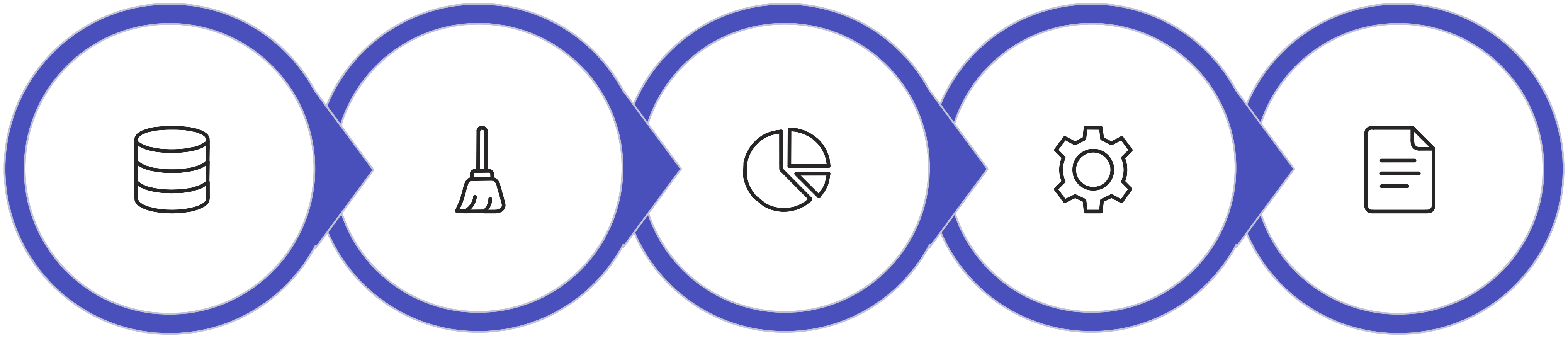
Jupyter / VS Code

Interactive notebooks for exploration and VS Code for reproducible scripts and debugging.

Git

Version control and collaborative code management for notebooks and scripts.

Methodology & Workflow



Data Collection

Cleaning

Exploratory Analysis

Model Train/Test

Results & Report

Workflow summary: ingest raw/trained datasets using Pandas, apply deterministic cleaning rules and feature engineering, explore patterns with visual analytics, run trained ML models with Scikit-learn and produce concise reports for engineering review.

KEY WORK UNDERTAKEN

Key Work Undertaken

1. Orientation & Infrastructure

Studied plant organisation, production systems and IT data pipelines.

2. Hands-on Python Training

Fundamentals, libraries and scripting patterns for industrial datasets.

3. Dataset Analysis

Examined feature sets, target variables and data provenance for sample datasets.

4. Data Cleaning

Addressed missing values, outliers and inconsistencies using Pandas pipelines.

5. Exploratory Analysis

Identified trends, correlations and distributions; prepared visual summaries.

6. Model Application & Evaluation

Deployed trained models, measured metrics and validated results against domain knowledge.

Outcomes Achieved

Technical Proficiency

Working knowledge of Python, Pandas, NumPy, Matplotlib and Scikit-learn for industrial data tasks.

Data Handling Experience

Practical experience cleaning, processing and preparing trained datasets for analysis.

Analytical Output

Produced visualisations and concise reports to communicate actionable insights to stakeholders.

Domain Understanding

Improved appreciation of plant-scale operations and how data reflects process behaviour.

Professional Growth

Enhanced documentation, teamwork and technical communication skills for academic and industry roles.

Challenges & Key Learnings

Challenges Faced

- Messy, inconsistent real-world data: missing values, mixed units and duplicates.
- Aligning model outputs with engineering context — domain validation required.
- Balancing model complexity with interpretability for operational stakeholders.

Key Learnings

- Data cleaning is essential — it consumes the majority of analysis time but yields the most value.
- Choose visualisations that match the data type and audience for clearer decision-making.
- Cross-functional collaboration and documentation improve the practical usefulness of ML outputs.

Next steps: refine model validation with domain experts, implement automated data pipelines and expand analyses to additional plant datasets.